

João Rietra

AI Engineer · generative AI, LLMs, RAG, agents and model evaluation

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SUMMARY

AI Engineer and Data Scientist with an innovative model evaluation standard for any domain. Tested in my Master's dissertation and replicated in every model I ship today. I measure calibration and out-of-distribution generalisation as acceptance criteria. I design the data architecture that underpins production, from the relational store to the vector store. I judge a project's maturity by pipeline robustness, not by a metric in a controlled environment.

KEYWORDS

Python · PyTorch · MLflow · Docker · AWS · Git · SQL · model calibration (ECE) · out-of-distribution evaluation (OOD) · LGPD and sensitive data · generative AI · LLMs · RAG · agentic workflows · tool-calling · Model Context Protocol (MCP) · prompt engineering · TypeScript · NestJS · React · PostgreSQL (pgvector) · Groq · FastAPI · Redis/BullMQ · CI/CD (GitHub Actions) · fine-tuning (LoRA) · technical mentoring and teaching · subproject technical leadership · cross-functional teamwork · agile methodologies (Scrum/Kanban)

EDUCATION

- **Professional Master's in Software Engineering** — CESAR School, Recife · Oct 2025 – defence expected Nov 2026
 - **Doctor of Medicine (MD)** — FPS / IMIP, Recife · completed 2022
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EXPERIENCE

AI Engineer — LetzGrow (School Innovation, CESAR School) · 2026 – present

AI-assisted career platform: candidates move through a guided journey with an AI mentor and receive a personalised development plan. Selected for 2026.2, as a two-person team.

- **AI mentor:** conversational agent on an LLM (Groq, Llama 3.3 70B) that runs the journey and generates personalised track recommendations.
- **Agentic workflows:** development-plan generation with adaptive micro-task decomposition instead of a fixed template.
- **RAG:** recommendations grounded in the user profile and the domain through pgvector semantic search.
- **NestJS 11 + Prisma 5 backend on PostgreSQL 15,** Swagger-documented API, JWT authentication (access + refresh with rotation) and LinkedIn OAuth.
- **Asynchronous GitHub profile ingestion** with BullMQ + Redis, processed outside the request with retries.
- **AWS CDK and CI/CD on GitHub Actions:** automatic deploy on merge to `main`, with an OIDC role and secrets in Secrets Manager; conditional access and audit trail aligned with Brazil's LGPD (GDPR-equivalent).

Technical Lead (data and computer vision) — MAPEN (School Innovation, CESAR School) · 2026 – present

System that locates energy losses by comparing distributed against consumed energy, with red flags by region. Selected for 2026.2.

- **Nationwide data base** (hourly grid-operator series 2000→2026, billed consumption by state and IBGE boundaries in DuckDB) and automatic meter reading by fine-tuning TrOCR-small, tracked in MLflow.

ML Engineer — Veritas (School Innovation, CESAR School) · 2025 – 2026

Deepfake detection.

- **Trust triad:** CNN forensic score, Grad-CAM heatmap and contextual reading by a generative model (Gemini/GPT) in the same API.
- **FastAPI + SQLAlchemy backend** that loads the model in the lifespan and exposes a prediction endpoint for image uploads.

Software Engineer (Clinical Domain + QA) — GasoGraph (Master's / CESAR) · 2025 – 2026

Clinical decision support system (CDSS) for arterial blood-gas interpretation in veterinary care, with rules from published consensus.

- **Clinical decision graphs and rule engine:** turned the consensus into flowcharts and implemented the rule tree from them.
- **Clinically driven QA:** tested every version the way a physician would use it, checking that the tool met its clinical purpose.

Teaching Assistant — CESAR School · 2025 – 2026

- Programming (C), Statistics (R), Calculus I, Competitive Programming and Data Analysis and Visualization.

Medical Data Analyst — J & L Hospital Consulting · 2023 – 2024

Analysis of clinical and managerial data to support hospital governance.

- **Clinical outcomes:** measured readmission, hospital-acquired infection and post-operative complications, pointing out failures and supporting protocol adjustments.
- **Quality and management:** tracked ONA and JCI accreditation KPIs and analysed bed occupancy, operating room and costs, with Streamlit dashboards.

PROJECTS

ragcode — own MCP server · github.com/jhlr/ragcode

MCP server (stdio) that exposes a local LLM (Ollama) as tools, with an incremental per-commit semantic code index in SQLite.

Minnow — coding agent in the phone browser · github.com/jhlr/minnow

Whole agent on the client: permissioned tools, IndexedDB, Pyodide, isomorphic-git and an on-device WebAssembly model with tool-calling; turn cut from 420 s to 176 s.

ACADEMIC OUTPUT

1. *Training Reuse for Binary Triage in Multi-Domain Medical Imaging* (Master's dissertation, CESAR School, 2026): 4 backbone origins across 5 domains, with calibration ECE 0.045 vs. 0.219 for ImageNet.
2. *Asymmetric generalisation in image-manipulation detectors: from the training domain to collapse on manual splicing* (SBTI 2026, submitted).
3. *Fine-tuning decision forests under drift: a Bayesian blend with a unified vocabulary* (SBTI 2026, submitted).
4. *EfficientNeXt: an autoregressive mini-LLM derived from the EfficientNet architecture with unified generation and embedding (GRIT) and decoding steering without an auxiliary model*.
5. *Automated inspection of electricity-meter reading photos: an OCR pipeline and characterisation of the coherence-validation problem* (SBTI 2026, submitted).

Papers written in Portuguese; original titles are in the Portuguese version of this CV.

SKILLS

Generative AI: OpenAI, Anthropic, Groq, Gemini and Ollama · prompt engineering · RAG and vector search · agentic workflows and tool-calling · Model Context Protocol (MCP) · LoRA fine-tuning · LLM evaluation (groundedness, hallucination)

Machine learning and engineering: PyTorch · Hugging Face Transformers · OpenCV · OCR · Grad-CAM · calibration (ECE) · OOD evaluation · Python (FastAPI) · TypeScript (NestJS, React) · PostgreSQL and SQL · Redis/BullMQ · JWT/OAuth · Docker · AWS (CDK) · GitHub Actions · MLflow · DuckDB · LGPD and sensitive data

Languages: Portuguese (native) · English (native) · Spanish (basic) · French (basic)